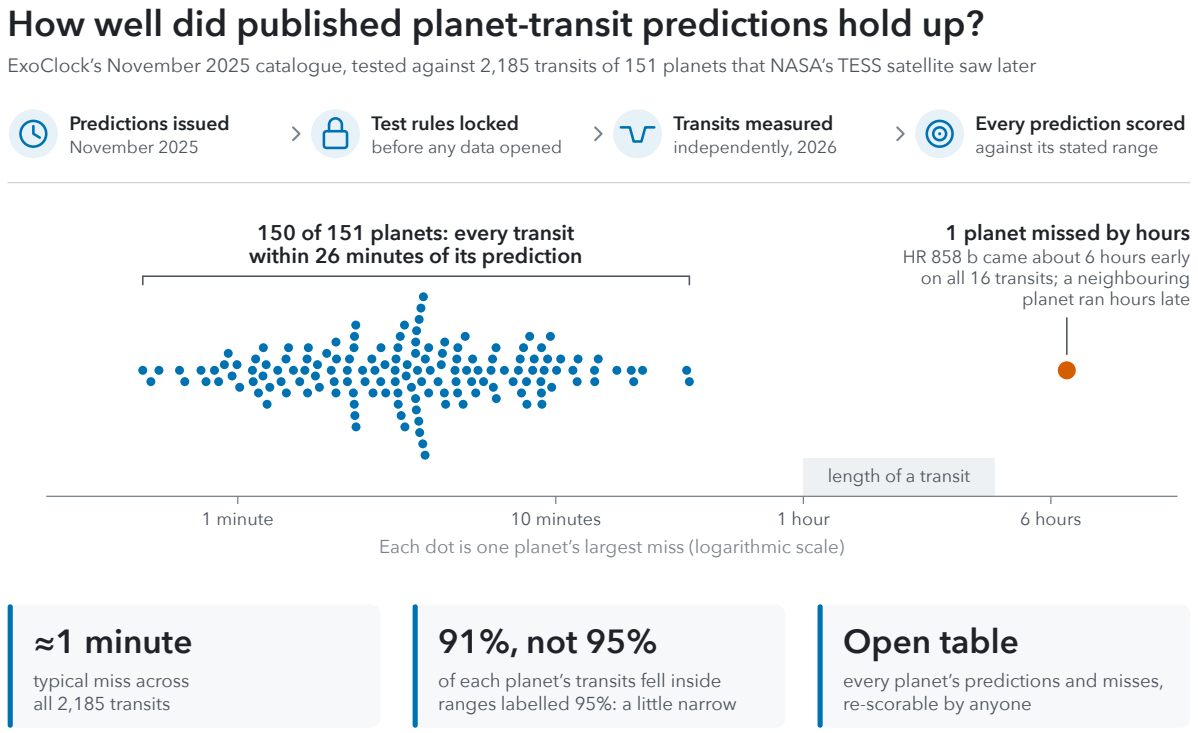
Predicted planet transits arrived within minutes, with one six-hour miss

Research brief · 12 November 2026 · aligned with the reconstructed article (18_final_draft.md). Codes in brackets are claim IDs in the project's claims ledger.



In one sentence: published predictions of when planets pass in front of their stars agreed with later measurements to minutes; one of 151 tested planets missed by hours.

Why it matters

Astronomers study many distant planets during a transit, the few hours in which a planet crosses the face of its star and starlight passing through its atmosphere can reveal what it is made of. Telescopes such as JWST, and the European Space Agency's coming Ariel mission, book observations from a predicted transit time and its stated uncertainty: too small, and the transit can be missed; too large, and scarce telescope time is wasted.

ExoClock, a project supporting Ariel, publishes these predictions in dated catalogue editions, each a forecast made before the transits it predicts. Its own check of the 2025 edition was partly optimistic, because some comparison timings had helped build it. We found no study that had fixed an edition in advance, measured later transits independently and scored its uncertainties.

What we did

We measured transit mid-times, the centre of each crossing, from later NASA TESS observations, with every rule locked before those data were opened (C042, C045). Artificial transits planted in real data showed that our ranges held the true times at about the stated rate, on the samples used to tune the method (C038). We scored each forecast for accuracy and range width, and tested a correction learned from the previous edition's errors. Our planet-by-planet table and scoring package let anyone test a better prediction against the same measurements (C053).

What we found

Accurate to minutes. We accepted 2,185 timings of 151 of the 156 planned planets. The typical miss was about a minute, and for 150 planets every transit came within 26.4 minutes of its prediction (C043, C063). The 2022 edition still held within 37 minutes nearly four years on, for every planet we could check (C062).

Ranges a little too narrow. Ranges labelled "95%", combined with our measurement uncertainty, contained about 91% of each planet's transits (C063). Misses were more often early than late, for reasons we have not identified (C065).

A correction did not clearly help. Widening the ranges 1.7-fold raised coverage above 95%, but the overall score could have improved or worsened substantially (C043). Of 151 planets, 111 scored worse; most of the average gain came from one planet, whose transits even the widened ranges never contained (C046, C059).

One six-hour miss. HR 858 b arrived about six hours early on all 16 measured transits, as two independently written programs confirm. Its neighbour, planet c, ran nine to ten hours late while planet d was on time: the pattern expected when planets tug on each other, and evidence against a shared clock shift (C048, C050, C051).

Warning signs, unproven. HR 858 b was new to the edition, its prediction rested on poorly fitting observations that ended in 2020, and its timing variations had been published a year earlier (C048, C054). But one case cannot validate a warning rule, and on the 2022 edition no tested scheme for giving selected planets wider windows beat both ways of spreading the same extra time evenly (C061, C064). This test, the four-year comparison and the early/late check were added after the data had been inspected.

What it means

For nearly every planet tested, an ExoClock prediction agreed with the measured transit to minutes, but the rare exception can be large. Warning information such as old inputs, a poor fit or published timing variations could be printed beside each prediction and tested in future editions. Publishing how the uncertainties of a reference transit time and the period are linked would make such tests more exact; we had to estimate that link (C054, C063).

What we can and cannot say

- **Only measured transits count:** 70% of those predicted, and 92% of those TESS was watching at the predicted time (C046). A transit far off schedule could be lost or assigned to the wrong orbit, so no catalogue-wide failure rate follows, and a mid-time does not guarantee capturing a whole transit.
- **One test:** one edition over its first year, on stars with one listed planet; multi-planet stars gave too few timings to judge (C044). Our cross-checks re-used the same TESS data, and one measurement run was repeated after a file-path fault, preserving every earlier result (C045).

The record

Code, results, claims ledger, reviews and author correspondence are in the public repository; the table and scoring package will get a DOI on release.

Sources: ExoClock catalogue editions on OSF; NASA TESS data via MAST; HR 858 timing variations from Bonfanti et al. 2025, doi:10.1051/0004-6361/202451300.

Barry W. Brook set the question, authorised the opening of the protected data and reviews release versions. Claude Opus 5.5 (Anthropic) and Codex GPT-6.1-sol (OpenAI) designed and ran the analyses, each built an independent timing check, and reviewed each other's work. Claude drafted this brief and its graphical abstract; Codex cross-checked the text. Code: MIT. Text: CC BY-SA 4.0.